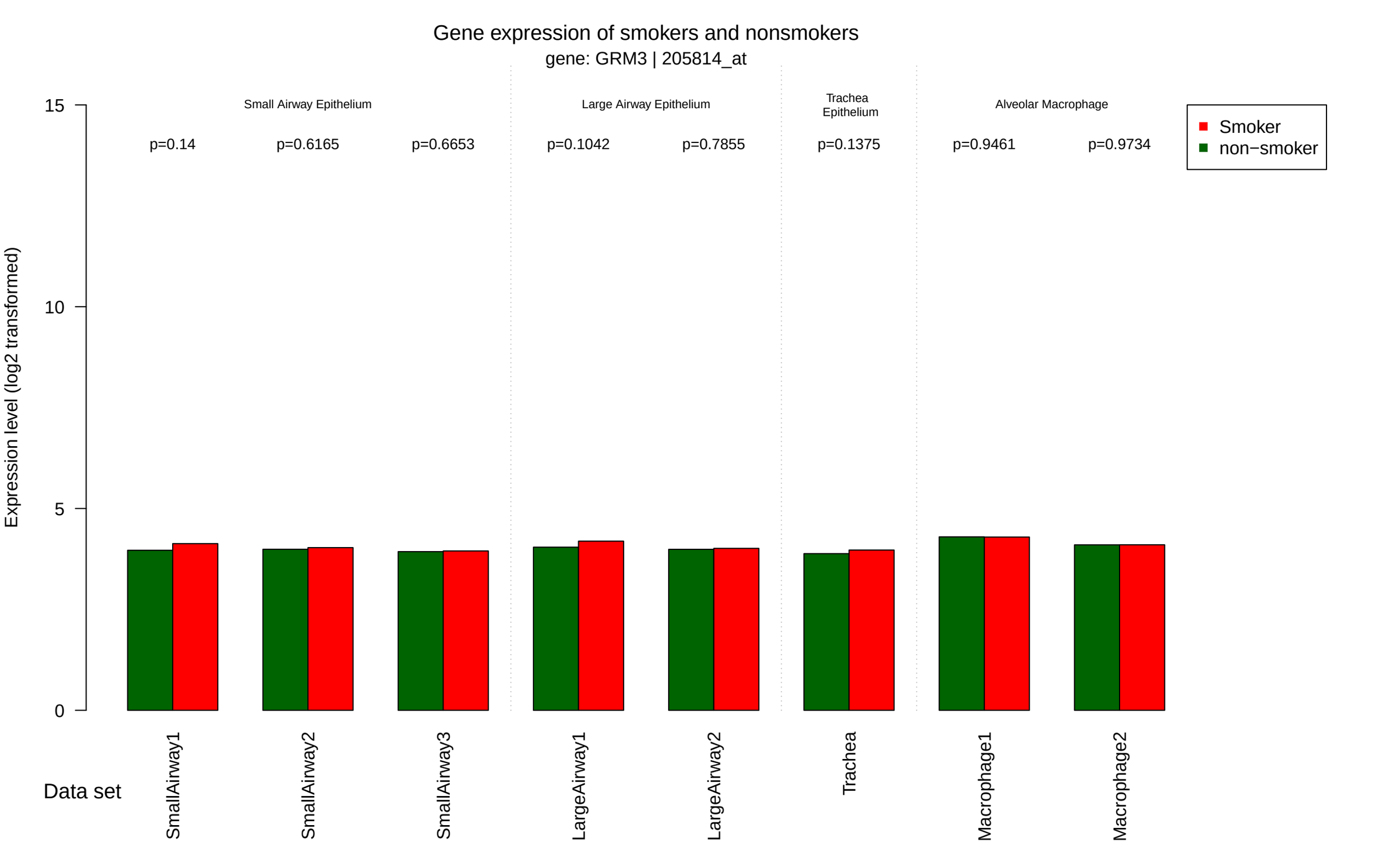
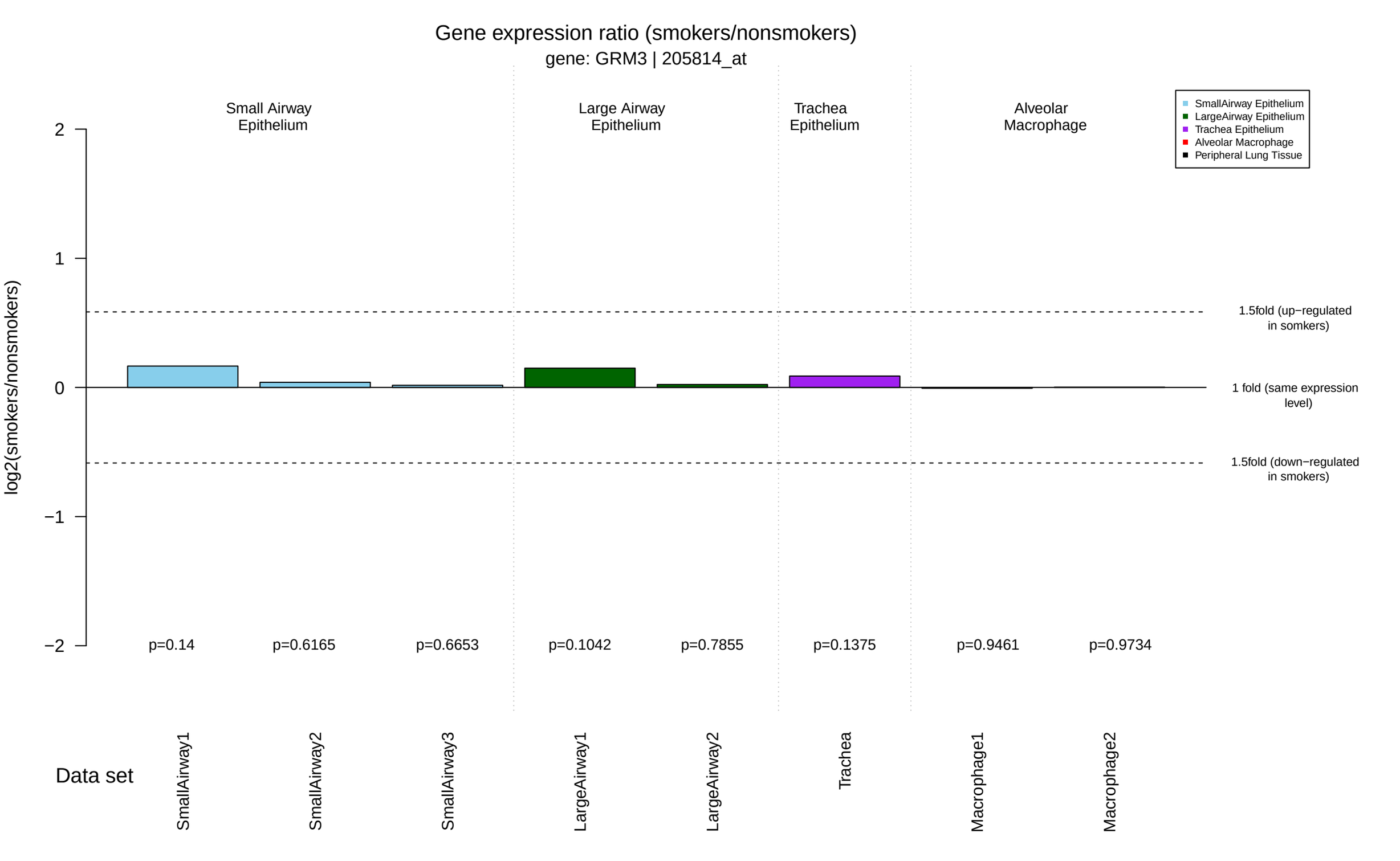
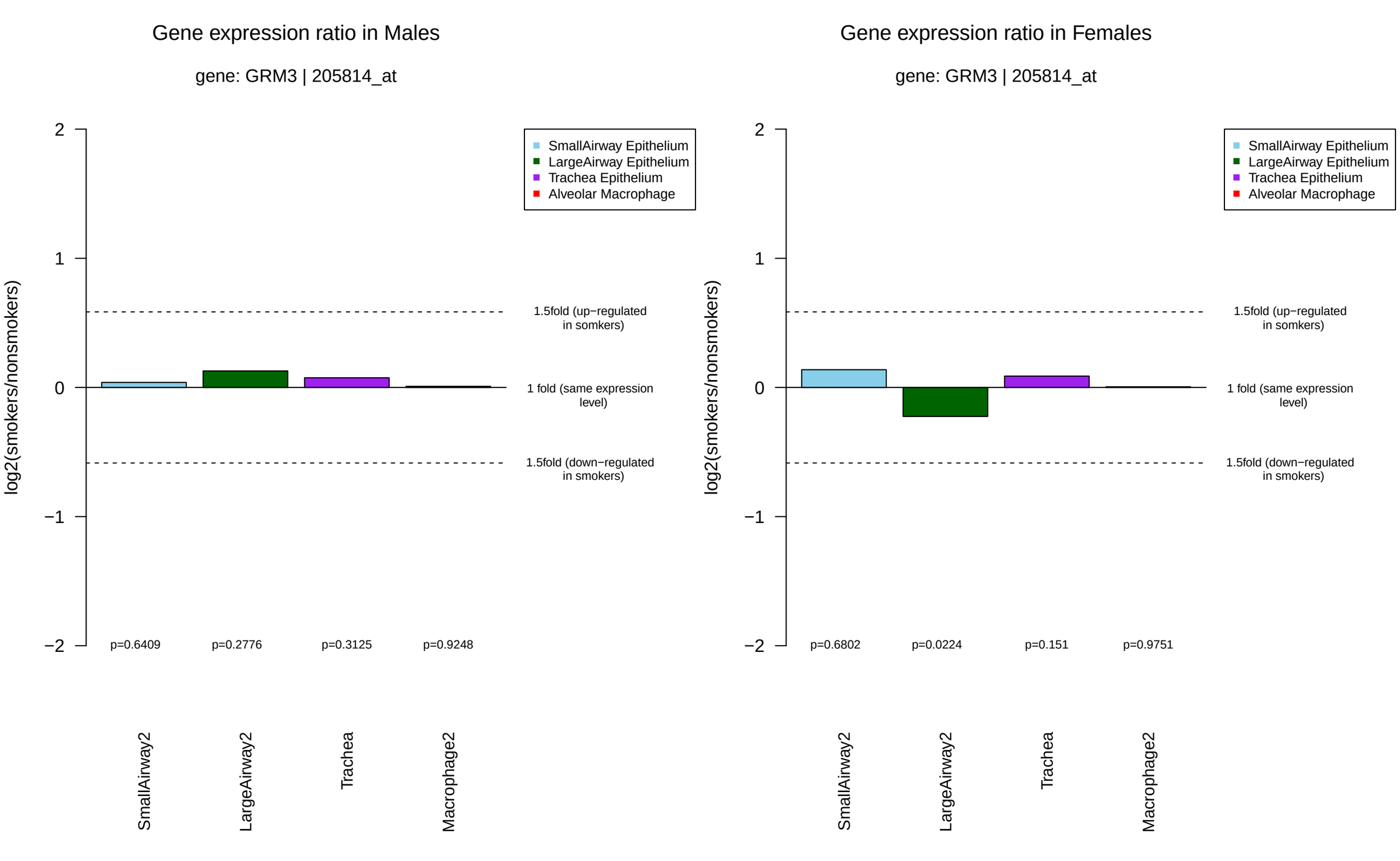
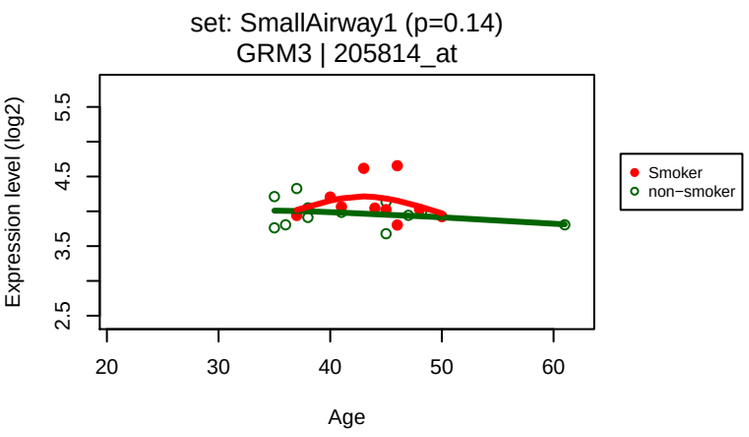
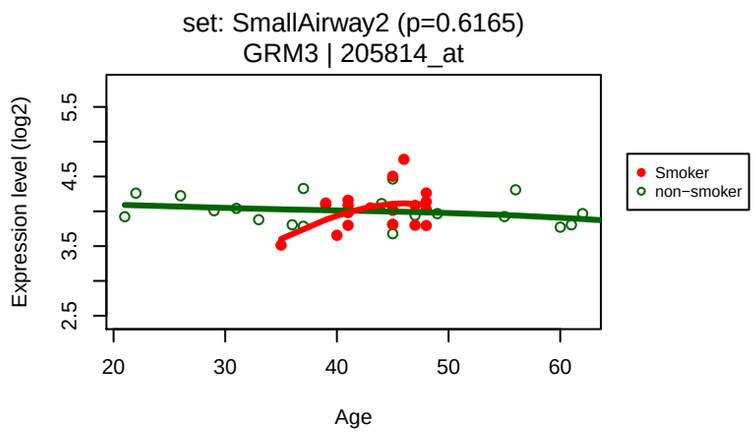
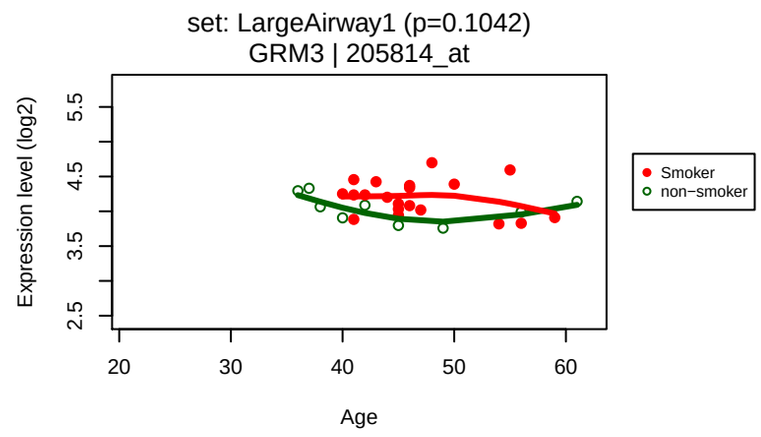

Gene expression of smokers and nonsmokers
gene: GRM3 | 205814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.14
p=0.6165
p=0.6653
p=0.1042
p=0.7855
p=0.1375
p=0.9461
p=0.9734
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

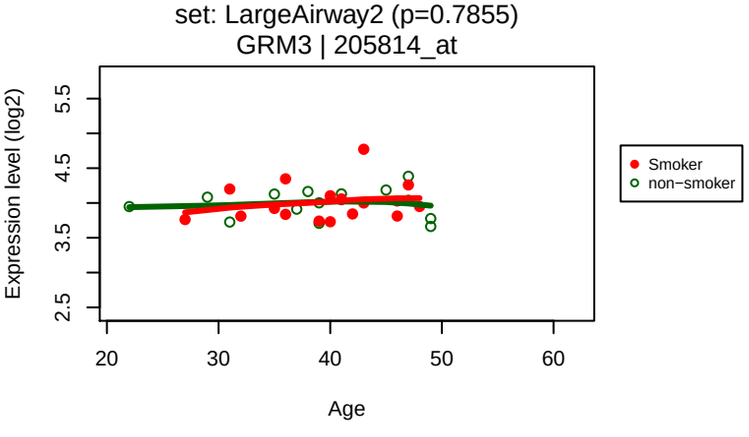

Gene expression ratio (smokers/nonsmokers)
gene: GRM3 | 205814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.14
p=0.6165
p=0.6653
p=0.1042
p=0.7855
p=0.1375
p=0.9461
p=0.9734
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

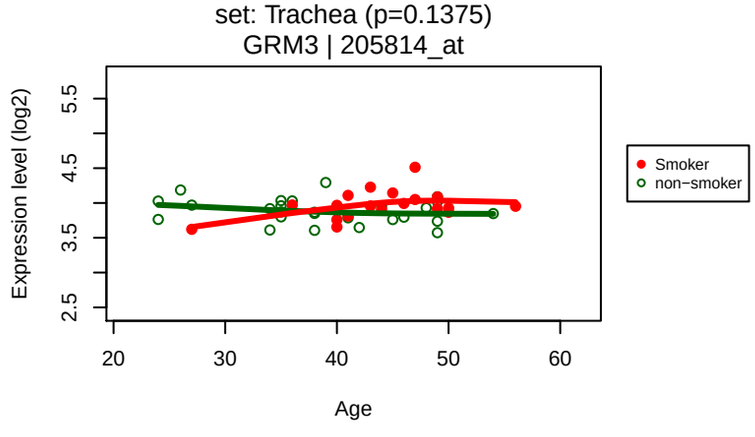

Gene expression ratio in Males
gene: GRM3 | 205814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6409
p=0.2776
p=0.3125
p=0.9248
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM3 | 205814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6802
p=0.0224
p=0.151
p=0.9751
SmallAirway2
LargeAirway2
Trachea
Macrophage2

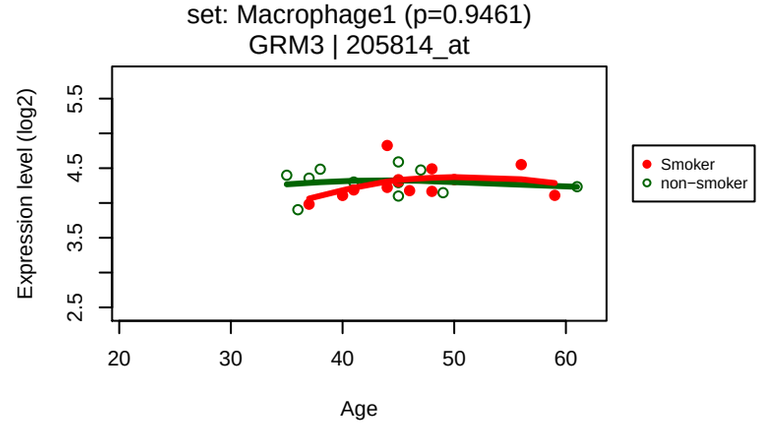

set: SmallAirway1 (p=0.14)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker

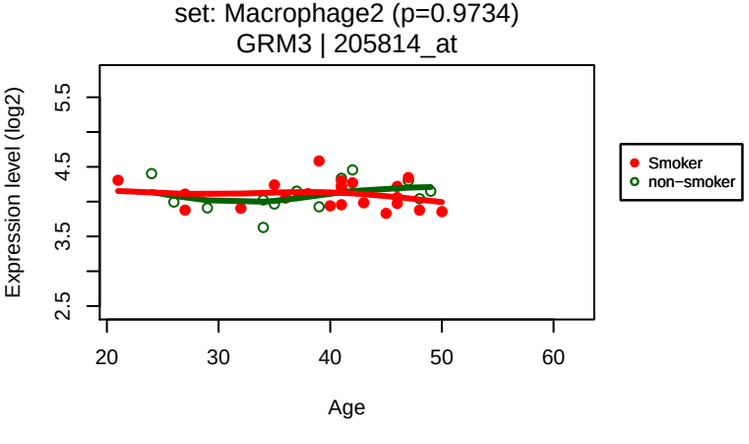

set: SmallAirway2 (p=0.6165)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker

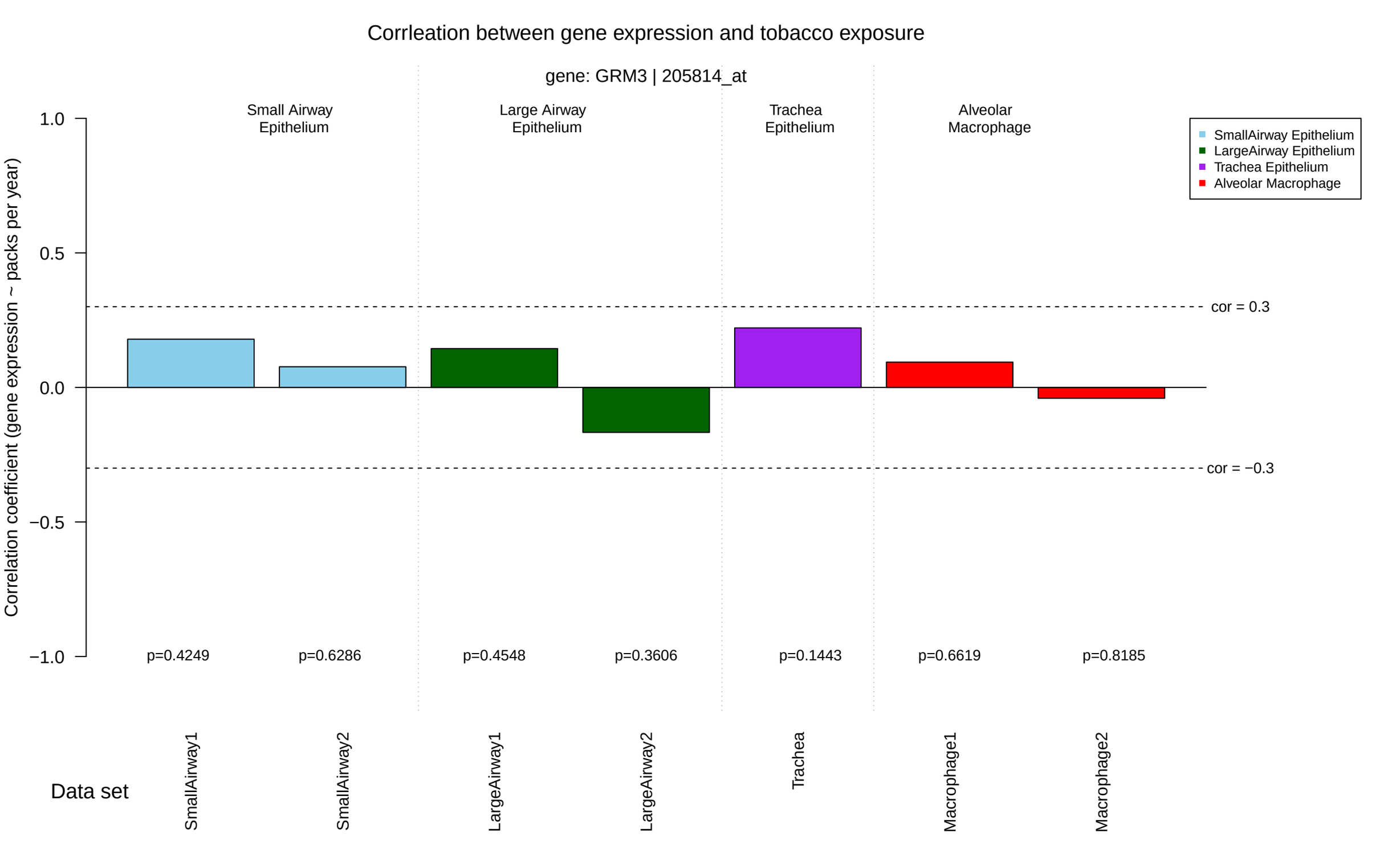

set: LargeAirway1 (p=0.1042)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7855)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1375)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9461)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9734)
GRM3 | 205814_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM3 | 205814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4249
p=0.6286
p=0.4548
p=0.3606
p=0.1443
p=0.6619
p=0.8185
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2